CSCE 679 DATA VISUALIZATION - PROJECT PROPOSAL

Brazos HealthLens: Linked, Uncertainty Aware Views of Tract Level Chronic Disease in the Brazos Valley

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1 Research Background and Motivation

Domain. Brazos HealthLens addresses *community health assessment*: describing how chronic disease, health behaviors and preventive care use vary between neighborhoods, so health departments and hospitals can decide where to act. Chronic diseases account for 7 of the 10 leading causes of death in the US [1], Texas had the highest uninsured rate of any state in 2024 (16.7%) [2], and every nonprofit hospital must complete a Community Health Needs Assessment (CHNA) every three years [3]; local health departments run similar assessments.

Why this domain. CDC PLACES now publishes model based estimates for every US census tract [1, 4], but its interactive map shows one measure at a time [5]. Multivariate questions such as "which tracts combine high diabetes, few checkups and many uninsured adults, and which places elsewhere in Texas look like ours?" still require toggling layers and exporting spreadsheets, and confidence intervals are easy to miss. We focus on the seven county Brazos Valley (Brazos, Burleson, Grimes, Leon, Madison, Robertson, Washington; 102 tracts with estimates), where a young, university driven core (Bryan–College Station) sits beside small town and rural counties: 76 valley tracts are metropolitan, 9 micropolitan and 17 rural by USDA RUCA codes [6]. Health profiles change sharply within a short drive. In Brazos County alone, crude diabetes prevalence ranges from 2.8% [95% CI 2.4–3.3] in College Station tract 21 (14,839 adults) to 23.7% [21.2–26.4] in Bryan tract 5.02 (976 adults). Part of that gap is age structure, since tract estimates are crude rather than age adjusted [7], and the smaller tract's interval is almost six times wider. Showing such differences together with their context and reliability is our design problem, and because our team is local, practitioners in the region are reachable.

Research question. How can coordinated, uncertainty aware visualizations help public health practitioners discover, compare and interpret multivariate chronic disease patterns across communities?

2 Design Requirements

2.1 Target users

Primary users are epidemiologists and program planners at the Brazos County Health District and hospital community health staff who prepare CHNAs; they decide where programs go and must justify those choices. **Secondary users** are clinicians planning outreach (e.g., hypertension follow up), regional planners such as Brazos Valley Council of Governments staff, and Texas A&M School of Public Health faculty and students. All are domain experts but not visualization experts, and none know 11 digit tract IDs, so every view needs place name search, readable labels and visible data caveats.

2.2 How we interact with target users

No user studies have been conducted yet; this is our plan. Because an early prototype exists (Fig. 1), users can react to concrete views rather than sketches. The requirements below are hypotheses that each phase can confirm, reword or drop, and every final requirement will cite its evidence (e.g., "Q7; S4, S11 → R3", where S is a respondent ID), following design study practice [8].

(1) **Formative survey, Oct 12–25.** After asking the instructor whether IRB review is needed and piloting with two classmates, we send a 10 minute anonymous online survey (target 15–25 responses) to health district and hospital staff by direct email and to School of Public Health mailing lists. It asks which decisions respondents make with community health data, which measures they use together, how they compare communities, how much they trust tract estimates, and what frustrates them in current tools. Results are analyzed by role so student answers do not drown out practitioners.

(2) **Think aloud feedback, Nov 2–8.** 30 minute sessions (in person or by video) with phase 1 volunteers: a short walkthrough, then open exploration of their own area. We note which views are used, misread or ignored; people who cannot meet get a short screenshot survey.

(3) **Evaluation, Nov 17–24.** Task based sessions on tasks T1–T5 (accuracy, time, confidence), the System Usability Scale (target above 68) and, for two tasks, the CDC PLACES map as a baseline. Two teammates independently code open ended answers into an affinity diagram.

2.3 Design requirements and visualization tasks

We propose five requirements (R), each mapped to an abstract task (T) from Brehmer and Munzner's typology [9] and to the views that support it (Table 1). Ease of use is a design principle across all views rather than a separate requirement.

	Requirement: users need to…	Visualization task (T)	Supporting view(s)
R1	**Explore where.** See any measure across 102 valley and 6,844 Texas tracts; find places by name	T1 Locate; explore spatial distribution	Map (choropleth, place search, scope toggle)
R2	**Compare a tract's full profile.** Many measures at once, against its county, the valley and Texas; relate risk factors to outcomes	T2 Compare to references; explore multivariate patterns; identify correlations	Tract profile · Parallel coordinates · Scatterplot
R3	**Find similar and unusual tracts.** Similar health profiles anywhere in Texas; tracts that stand out from their group	T3 Find similar items; find clusters and outliers	Peer finder · Map colored by health profile group
R4	**Judge reliability and context.** How reliable each estimate is; whether urban–rural or age structure explains a difference	T4 Assess uncertainty; compare urban–rural groups	CI whiskers in all views · CI width map · Urban–rural filter
R5	**Find care gaps.** Where high disease burden meets weak preventive care, to target outreach	T5 Find extremes on two derived attributes	Care gap bivariate map

Table 1. Requirements → tasks → views. All views share one global selection and brush.

2.4 Data and abstraction

Sources. (i) CDC PLACES 2025 census tract release (dataset cwsq ngmh [4]; BRFSS 2023, five measures from 2022): all 33 measures published for Texas (225,846 rows, 6,844 tracts), each with a 95% CI and adult population. Values are model based crude prevalence, not age adjusted [7]; the release has no social need or chronic kidney disease measures for Texas, so we make no claims about either. (ii) U.S. Census Bureau 2020 cartographic boundaries [10]: 6,884 tracts, 40 without estimates drawn gray, simplified and stored as TopoJSON. (iii) USDA ERS 2020 RUCA codes [6], grouped as metropolitan, micropolitan, small town and rural.

Abstraction. Items are tracts with quantitative attributes (prevalences, CI bounds, population), categorical attributes (county, RUCA class, cluster) and polygon geometry, in the hierarchy tract → county → Brazos Valley → Texas.

Derived data (precomputed in Python): 23 core measures z scored across Texas tracts and signed so + = worse; PCA (three components reach 90% of variance) and k means on those components (k = 5, best silhouette for k = 5–7, stable across restarts); each tract's 20 nearest peers by Euclidean distance on the z scores; a burden index (diabetes, high BP, heart disease, stroke, COPD) and a care deficit index (uninsured, no checkup, no BP medicine, no cholesterol screening); and adult weighted county, valley, RUCA and Texas averages.

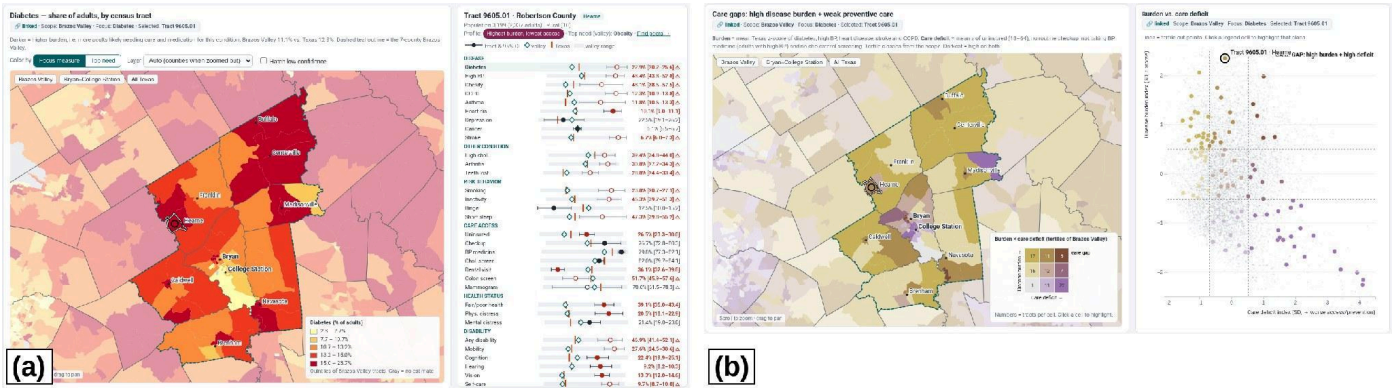


Figure 1. Early prototype, Brazos Valley scope. (a) Map and profile: diabetes choropleth with a selected Hearne tract and its dot and whisker profile (→ R1, R2, R4). (b) Care gap bivariate map and burden–deficit scatter (→ R5).

2.5 Views, encoding and interaction

Six linked views (map, tract profile, parallel coordinates, scatterplot, peer finder, care gap map) share one selection model: the selected tract is outlined in black and any brushed group is drawn in one highlight color in every view, i.e., global coordinated brushing [11, 12]. A scope toggle (Brazos Valley / All Texas) sets the reference population for color classes and rankings. Choropleths use five quantile classes, which suit epidemiological map series [13], with care measures reversed so darker always means worse; the care gap map uses a 3×3 bivariate scheme [14]; clusters use a colorblind safe palette. Uncertainty appears wherever values do: CI whiskers in the profile and peer panels, a CI width map mode, and hatching for low confidence estimates (CI width among the widest 10% for that measure). On screen notes warn that PLACES measures share model inputs and that tract associations do not describe individuals.

2.6 Early observations and usage scenario

Illustrative; to be replaced by scenarios gathered from users. A health district planner preparing hypertension outreach sets the focus measure to high blood pressure. The map points to Hearne tract 9605.01 in Robertson County, the valley's highest at 48.4% [43.8–52.8] (T1). It is hatched as low confidence, but even its lower bound is well above the valley (31.7%) and Texas (33.1%) averages (T2, T4). The peer finder lists similar tracts elsewhere in Texas, including Marlin, Jasper, Beaumont and Port Arthur, where she can look for programs that worked (T3). In the care gap view the tract has high burden but only a middle care deficit, so disease management may fit better than insurance enrollment; enrollment outreach points instead to five high burden, high deficit tracts in Bryan and Grimes (T5). Other early patterns: obesity and diabetes correlate at r = 0.80 across Texas tracts, and 61.6% of Brazos County adults with high blood pressure take medication versus 73.3% statewide. These are modeled estimates, so we treat them as prompts for discussion with users, not findings.

3 Related Work

Community health data platforms. CDC PLACES produces multilevel regression and poststratification estimates for counties, places, tracts and ZCTAs [1]; its map shows one measure at a time, and its comparison report covers up to three counties, not tracts. County Health Rankings ranks counties within each state [15]; with CDC's Community Health Status Indicators it offers peer counties found by k means on demographic and socioeconomic variables that deliberately exclude health outcomes [16]. The CDC/ATSDR Social Vulnerability Index maps census based vulnerability [17, 18], and the Dartmouth Atlas maps regional variation in Medicare care use [19]. Preim and Lawonn survey visual analytics for public health more broadly [20].

Coordinated and multivariate views. Brushing began with linked scatterplots [11], and Roberts reviews coordinated multiple views [12]. Shneiderman's "overview first, zoom and filter, then details on demand" [21] shapes our scope toggle and tract profile. Parallel coordinates [22] and small multiples [23] are established multivariate displays, and VIS STAMP linked maps, parallel coordinates and clustering to find multivariate spatial patterns [24]. Our design follows Munzner's what–why–how framework [25] and is implemented in D3 [26].

Maps and uncertainty. Quantile classes are among the most accurate for epidemiological map series [13], Trumbo set out principles for bivariate color maps [14], MacEachren et al. review geospatial uncertainty visualization [27], and value suppressing uncertainty palettes [28] are a candidate for a future single map of estimate and reliability.

Gap and potential novelty. Existing tools work at county level or one measure at a time, define peers without health data, and seldom show uncertainty. Brazos HealthLens combines, for one region at tract level: (1) six linked views over all 33 PLACES measures with one global brush; (2) health profile peer search across all 6,844 Texas tracts; (3) uncertainty shown in every view rather than a separate layer; (4) a care gap bivariate map for targeting outreach; (5) an urban–rural (RUCA) lens that keeps crude rate differences in context; and (6) requirements validated with local practitioners, which, as a design study, is itself part of our contribution.

4 Development Plans

Stack. Python (pandas, shapely, scikit learn) for preprocessing; TopoJSON boundaries; D3.js v7 with canvas rendering for dense views; one static offline HTML file, also hosted on GitHub Pages, with no backend. Each member owns one area (Table 2); all four share recruiting, running study sessions, code review, the report and the presentation.

Team member	Area	Responsibilities
Manogna Rayala	Multivariate views & integration	Parallel coordinates, scatterplot, global linking; merges code into one build; final demo
Alexis Santiago	User studies & evaluation	Survey design, feedback sessions, SUS and task analysis; requirement traceability
Diviya Bhavaani M B	Data pipeline	PLACES, boundaries and RUCA prep; clusters, peers, care gap index; methods and data documentation
Seo Hyeon Beom	Maps & uncertainty	Map, tract profile, uncertainty views; accessibility (keyboard, contrast); related work

Table 2. Task assignments. Each member has an equal share of ownership.

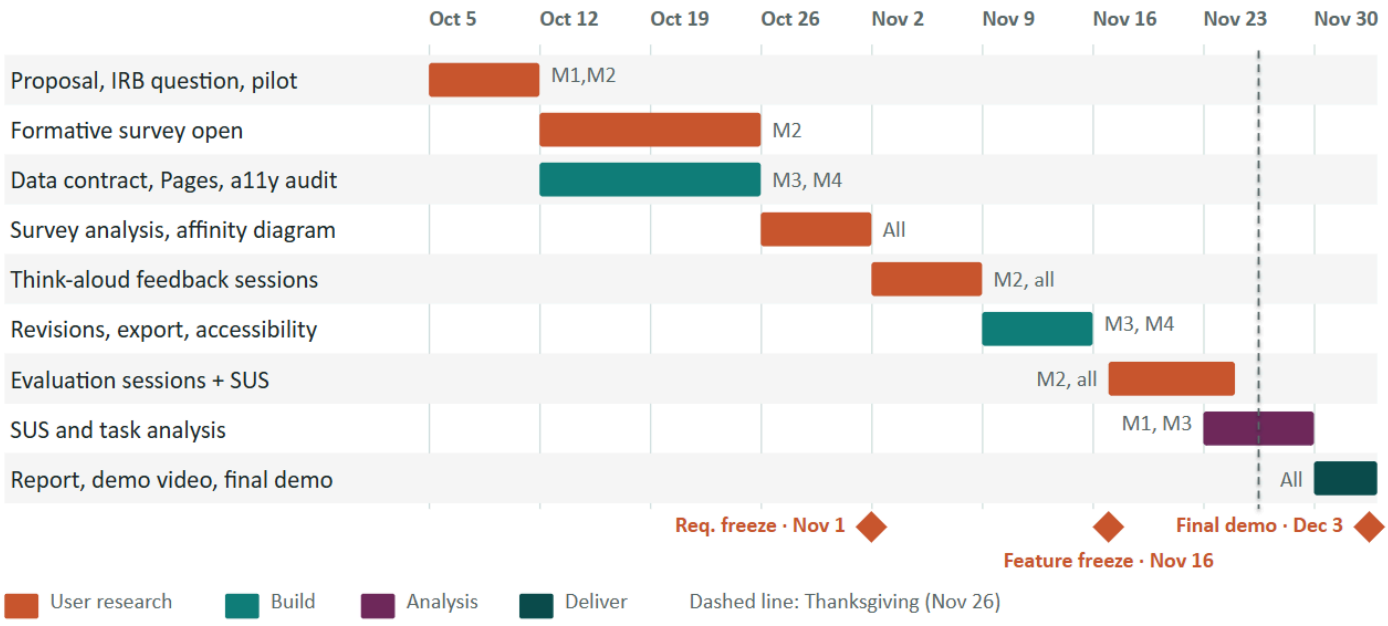


Figure 2. Timeline, Oct 5 – Dec 3, 2026. M1 = Manogna Rayala, M2 = Alexis Santiago, M3 = Diviya Bhavaani M B, M4 = Seo Hyeon Beom. Milestones: requirements freeze Nov 1, feature freeze Nov 16, final demo Dec 3. Fallback if survey responses are few: widen distribution and rely more on the feedback sessions.

5 Reflection on Using AI

How we used it. An AI assistant helped us build the early D3 prototype (from a first map to linked views), debug coordination (a scope switch that updated only one tab led us to a global state store and tests), write the PLACES, Census and RUCA data scripts, and draft and format the proposal, slides and this report, including finding candidate references.

What we had to fix. Every output needed checking against the data and sources. A draft claimed a College Station tract had the highest depression estimate in Texas; the data says third. A story title said four clusters; the data has five. Drafts misstated the PLACES release (it has 33 measures for Texas, crude rather than age adjusted). Some suggested citations could not be verified and one DOI did not resolve, so we dropped them. AI assisted drafts also leaned toward breadth: our first plan had ten requirements and twelve views, which we consolidated to five and six after instructor feedback.

Going forward. We use AI for scaffolding, tests and writing help, while the team owns design decisions and reviews and tests every change. We will not use AI generated user data, and we disclose AI use in the report and README.

References

[1]K. J. Greenlund, H. Lu, Y. Wang, K. A. Matthews, J. M. LeClercq, B. Lee, and S. A. Carlson, "PLACES: Local Data for Better Health," Preventing Chronic Disease, vol. 19, Art. 210459, 2022. doi:10.5888/pcd19.210459

[2]C. Carter, "Health Insurance Coverage by State: 2023 and 2024," American Community Survey Briefs ACSBR 024, U.S. Census Bureau, Sept. 2025. https://www2.census.gov/library/publications/2025/demo/acsbr 024.pdf

[3]Internal Revenue Service, "Community health needs assessment for charitable hospital organizations – Section 501(r)(3)." https://www.irs.gov/charities non profits/community health needs assessment for charitable hospital organizations section 501r3 (accessed Oct. 2, 2026)

[4]Centers for Disease Control and Prevention, "PLACES: Local Data for Better Health, Census Tract Data, 2025 release," dataset cwsq ngmh. https://data.cdc.gov/500 Cities Places/PLACES Local Data for Better Health Census Tract D/cwsq ngmh (retrieved Oct. 2, 2026)

[5]Centers for Disease Control and Prevention, "Interactive PLACES Map." https://www.cdc.gov/places/tools/interactive map tool.html (accessed Oct. 2, 2026)

[6]U.S. Department of Agriculture, Economic Research Service, "Rural Urban Commuting Area Codes" (2020 census tract codes). https://www.ers.usda.gov/data products/rural urban commuting area codes (downloaded Oct. 2, 2026)

[7]Centers for Disease Control and Prevention, "PLACES Frequently Asked Questions." https://www.cdc.gov/places/faqs/index.html (accessed Oct. 2, 2026)

[8]M. Sedlmair, M. Meyer, and T. Munzner, "Design Study Methodology: Reflections from the Trenches and the Stacks," IEEE Trans. Visualization and Computer Graphics, vol. 18, no. 12, pp. 2431–2440, 2012. doi:10.1109/TVCG.2012.213

[9]M. Brehmer and T. Munzner, "A Multi Level Typology of Abstract Visualization Tasks," IEEE Trans. Visualization and Computer Graphics, vol. 19, no. 12, pp. 2376–2385, 2013. doi:10.1109/TVCG.2013.124

[10]U.S. Census Bureau, "Cartographic Boundary Files" (2020; cb_2020_48_tract_500k and cb_2020_48_place_500k). https://www.census.gov/geographies/mapping files/time series/geo/cartographic boundary.html

[11]R. A. Becker and W. S. Cleveland, "Brushing Scatterplots," Technometrics, vol. 29, no. 2, pp. 127–142, 1987. doi:10.1080/00401706.1987.10488204

[12]J. C. Roberts, "State of the Art: Coordinated & Multiple Views in Exploratory Visualization," in Proc. 5th Int. Conf. Coordinated and Multiple Views in Exploratory Visualization (CMV 2007), pp. 61–71, 2007. doi:10.1109/CMV.2007.20

[13]C. A. Brewer and L. Pickle, "Evaluation of Methods for Classifying Epidemiological Data on Choropleth Maps in Series," Annals of the Association of American Geographers, vol. 92, no. 4, pp. 662–681, 2002. doi:10.1111/1467 8306.00310

[14]B. E. Trumbo, "A Theory for Coloring Bivariate Statistical Maps," The American Statistician, vol. 35, no. 4, pp. 220–226, 1981. doi:10.1080/00031305.1981.10479360

[15]P. L. Remington, B. B. Catlin, and K. P. Gennuso, "The County Health Rankings: rationale and methods," Population Health Metrics, vol. 13, Art. 11, 2015. doi:10.1186/s12963 015 0044 2

[16]County Health Rankings & Roadmaps, "Peer County Methodology Used by the Community Health Status Indicators 2015 Web Application." https://www.countyhealthrankings.org/sites/default/files/media/document/resources/CHSIpeerMethodology.pdf (accessed Oct. 2, 2026)

[17]B. E. Flanagan, E. W. Gregory, E. J. Hallisey, J. L. Heitgerd, and B. Lewis, "A Social Vulnerability Index for Disaster Management," J. Homeland Security and Emergency Management, vol. 8, no. 1, 2011. doi:10.2202/1547 7355.1792

[18]Agency for Toxic Substances and Disease Registry / CDC, "CDC/ATSDR Social Vulnerability Index." https://www.atsdr.cdc.gov/place health/php/svi/index.html (accessed Oct. 2, 2026)

[19]The Dartmouth Atlas of Health Care. https://www.dartmouthatlas.org/ (accessed Oct. 2, 2026)

[20]B. Preim and K. Lawonn, "A Survey of Visual Analytics for Public Health," Computer Graphics Forum, vol. 39, no. 1, pp. 543–580, 2020. doi:10.1111/cgf.13891

[21]B. Shneiderman, "The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations," in Proc. IEEE Symp. Visual Languages, pp. 336–343, 1996. doi:10.1109/VL.1996.545307

[22]A. Inselberg, "The plane with parallel coordinates," The Visual Computer, vol. 1, no. 2, pp. 69–91, 1985. doi:10.1007/BF01898350

[23]E. R. Tufte, Envisioning Information, ch. 4 "Small Multiples." Cheshire, CT: Graphics Press, 1990.

[24]D. Guo, J. Chen, A. M. MacEachren, and K. Liao, "A Visualization System for Space Time and Multivariate Patterns (VIS STAMP)," IEEE Trans. Visualization and Computer Graphics, vol. 12, no. 6, pp. 1461–1474, 2006. doi:10.1109/TVCG.2006.84

[25]T. Munzner, Visualization Analysis and Design. Boca Raton, FL: CRC Press (A K Peters), 2014. doi:10.1201/b17511

[26]M. Bostock, V. Ogievetsky, and J. Heer, "D³: Data Driven Documents," IEEE Trans. Visualization and Computer Graphics, vol. 17, no. 12, pp. 2301–2309, 2011. doi:10.1109/TVCG.2011.185

[27]A. M. MacEachren, A. Robinson, S. Hopper, S. Gardner, R. Murray, M. Gahegan, and E. Hetzler, "Visualizing Geospatial Information Uncertainty: What We Know and What We Need to Know," Cartography and Geographic Information Science, vol. 32, no. 3, pp. 139–160, 2005. doi:10.1559/1523040054738936

[28]M. Correll, D. Moritz, and J. Heer, "Value Suppressing Uncertainty Palettes," in Proc. 2018 CHI Conf. Human Factors in Computing Systems, pp. 1–11, 2018. doi:10.1145/3173574.3174216